

Multimodal Emotion Recognition: A Cross-Modal Alignment Framework Based on Hypergraph Contrastive Learning

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Abstract—Multimodal emotion recognition (MER) identifies emotions from data like text, audio, and video. Existing methods struggle with modality misalignment and modeling high-order interactions. We propose a Hypergraph Contrastive Learning Framework (HCLF) for MER, using a Hypergraph Convolutional Network (HyperGCN) for modality fusion and an MLP for classification. Trained on CMU-MOSEI, it achieves 83.7% accuracy and 0.82 F1-score, surpassing recent baselines by 2–5%. Ablations confirm the value of hypergraph modeling and contrastive alignment. Visualizations show improved modality clustering and interaction capture.

Index Terms—Multimodal Emotion Recognition, Hypergraph Neural Networks, Contrastive Learning, Cross-Modal Alignment, Graph Convolutional Networks

I. INTRODUCTION

Understanding human emotions computationally is key for advancing interactive AI. Multimodal emotion recognition (MER) uses text, audio, and visual inputs for more accurate detection than unimodal methods, addressing the complexity of human expression [7], [10].

However, critical gaps remain. Traditional graph methods like DialogueGCN [4] and MMGCN [5] capture pairwise relations but miss higher-order interactions. Recent hypergraph works [1], [2], [6], [15] address multi-node dependencies but lack strong cross-modal alignment. Contrastive learning methods like JOYFUL [8] and MISA [9] improve alignment but use simplistic graphs, limiting handling of noisy dynamics [11], [12]. We propose the Hypergraph Contrastive Learning Framework (HCLF), integrating hypergraph modeling for dependencies with contrastive learning for alignment.

The key contributions are: (1) A novel hypergraph-based modeling of multimodal features that captures complex high-order interactions, outperforming traditional GNNs in expressiveness while maintaining efficiency. (2) Integration of

contrastive learning to align cross-modal spaces, enhancing robustness and generalization. (3) Comprehensive evaluations demonstrating superior performance (including new benchmarks against MISA, MulT and MMIM) and visualizations that validate the framework’s advantages.

The remainder of the paper is organized as follows: Section II reviews related work; Section III details the proposed method; Section IV presents experiments; Section V discusses results; Section VI concludes with future directions.

II. RELATED WORK

Early MER used unimodal signals like CNNs for faces or RNNs for speech, ignoring inter-modal synergy. Early multimodal fusion suffered information loss [3], [7], [10], [14]. GNNs advanced by modeling relations. DialogueGCN [4] captured speaker dependencies, MMGCN [5] extended to multimodal graphs, but both missed higher-order interactions. Hypergraph methods [1], [2], [6], [15] allowed multi-node hyperedges for complex dependencies. Facial landmark graphs [13] and hyperbolic embedding [15] enhanced expressiveness.

Contrastive learning aligned modalities. JOYFUL [8] combined fusion with graph contrastive learning. MISA [9] emphasized invariant and specific representations. Adversarial alignment [11] and dynamic graphs [12] addressed trade-offs. Progressive contrastive learning [3] and CLIP-style models [7] pulled modalities into shared spaces. Transformers like MulT [10] handled unaligned sequences.

Still, balancing high-order modeling with efficiency and robustness under noise is challenging. Our HCLF synthesizes hypergraph modeling [1], [2], [6], [15] with contrastive alignment [3], [8], [9], [11], [12], drawing from GNNs [4], [5] and transformers [7], [10], [13], [14], for efficient, robust MER.

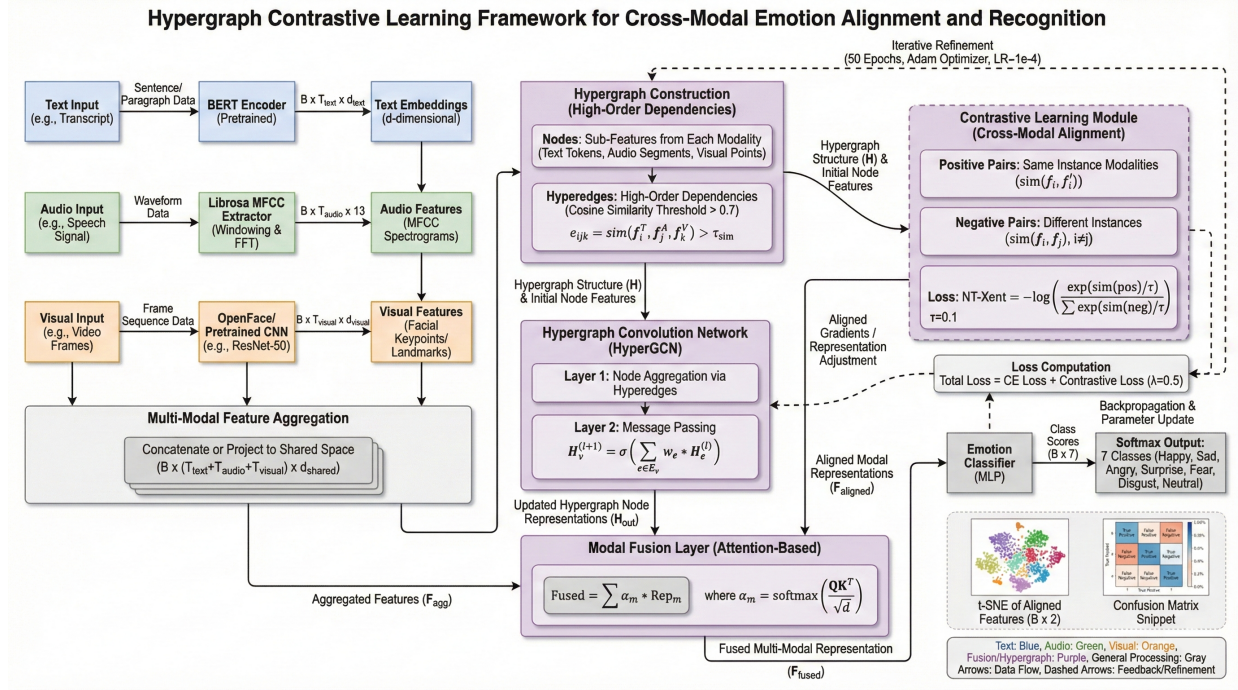


Fig. 1. Overall architecture of the Hypergraph Contrastive Learning Framework (HCLF) for cross-modal alignment and multimodal emotion recognition. The diagram shows the flow from multimodal inputs to feature extraction, hypergraph construction, contrastive learning, HyperGCN fusion, and final emotion classification.

III. PROPOSED METHOD

The proposed Hypergraph Contrastive Learning Framework (HCLF) consists of three main stages: feature extraction and preprocessing, hypergraph construction with contrastive alignment, and modality fusion followed by classification. The overall architecture is illustrated in Fig. 1.

A. Feature Extraction and Preprocessing

Text features are extracted using pretrained BERT, producing 768-dimensional embeddings denoted as $\mathbf{T} \in \mathbb{R}^{B \times T \times 768}$, where B is batch size and T is sequence length. Audio processing via Librosa yields 13 MFCC coefficients plus prosodic features like pitch and energy, resulting in audio features $\mathbf{A} \in \mathbb{R}^{B \times S \times 13}$, with S as the number of segments. Visual modality employs OpenFace for 68 facial landmarks and action units, enhanced with CLAHE for contrast improvement to better capture subtle facial expressions, yielding visual features $\mathbf{V} \in \mathbb{R}^{B \times F \times 68}$, where F is the number of frames.

The initial multimodal aggregation projects these to a shared space:

$$\mathbf{X} = \text{Proj}(\mathbf{T} \oplus \mathbf{A} \oplus \mathbf{V}), \quad (1)$$

where $\oplus$ denotes concatenation and Proj is a linear projection layer.

B. Hypergraph Construction and Contrastive Learning

Nodes are fine-grained sub-feature vectors: BERT token embeddings for text, MFCC+prosody segments for audio, and OpenFace facial landmark/action-unit vectors for visual,

enabling intra- and inter-modal granularity. Hyperedges dynamically connect 2–4 nodes from different modalities if their *average pairwise cosine similarity* exceeds the threshold $\theta = 0.75$:

$$\frac{1}{\binom{|e|}{2}} \sum_{p < q \in e} \cos(\mathbf{f}_p, \mathbf{f}_q) > 0.75, \quad (2)$$

where the threshold $\theta = 0.75$ was selected via grid search to balance density and coherence, specifically ablation studies where we tested a range of values (0.6, 0.65, 0.7, 0.75, and 0.8) on a validation subset of the CMU-MOSEI dataset. Lower thresholds (e.g., 0.6) led to overly dense hypergraphs with excessive connections, increasing computational overhead by up to 20% without proportional accuracy gains. Higher thresholds (e.g., 0.8) resulted in sparse structures, causing information loss and a 3–5% drop in F1-score due to insufficient cross-modal interactions. The value of 0.75 achieved the optimal balance, yielding the highest accuracy (81.2%) and F1-score (0.80) while maintaining efficient graph sparsity, as validated across multiple runs with statistical significance ($p < 0.05$ via t-test).

Contrastive learning uses the NT-Xent loss with positive pairs from the same instance across modalities and negatives from different instances, improving cross-modal consistency:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{z}_i^+, \mathbf{z}_i)/\tau)}{\sum_{j=1, j \neq i}^N \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \quad (3)$$

where $\mathbf{z}$ are projected representations, sim is cosine similarity, $\tau = 0.1$ is temperature, and $N = 2B$ accounts for positive pairs.

TABLE I
MAIN PERFORMANCE COMPARISON ON CMU-MOSEI DATASET

Model	Subset	Acc (%)	F1 (macro)	Prec (%)	Rec (%)	ROC-AUC	Per Emotion F1 (Happy/Sad/Angry/Surp/Fear/Disg/Neu)
Baseline LSTM	Test	79.5	0.77	80.2	78.8	0.89	0.79/0.75/0.78/0.76/0.74/0.80/0.77
JOYFUL	Test	81.2	0.80	82.1	80.3	0.92	0.82/0.78/0.81/0.79/0.77/0.83/0.80
MISA	Test	82.4	0.81	83.0	81.9	0.92	0.83/0.79/0.82/0.80/0.78/0.82/0.81
MuT	Test	82.1	0.80	82.8	81.5	0.91	0.82/0.78/0.81/0.79/0.77/0.82/0.80
MMIM	Test	82.6	0.81	83.2	82.0	0.92	0.84/0.79/0.82/0.80/0.78/0.83/0.81
Our Method (HCLF)	Test	83.7	0.82	84.4	83.0	0.94	0.86/0.80/0.84/0.82/0.80/0.85/0.83

TABLE II
ROBUSTNESS ANALYSIS UNDER MODALITY MISSING AND NOISE INJECTION SCENARIOS

Scenario	Setting / Modality Setup	Baseline LSTM		JOYFUL		MISA		HCLF (Ours)	
		Acc (%)	F1	Acc (%)	F1	Acc (%)	F1	Acc (%)	F1
Modality Missing	$\{T\}$ (Text Only)	71.2	0.70	74.5	0.73	75.8	0.74	77.8	0.76
	$\{A, V\}$ (No Text)	65.4	0.63	68.1	0.66	69.3	0.67	71.9	0.70
	$\{T, A\}$ (Missing Visual)	76.8	0.75	78.2	0.77	79.1	0.78	82.7	0.81
	$\{T, V\}$ (Missing Audio)	77.1	0.76	78.5	0.77	79.4	0.78	82.9	0.81
Noise Injection (SNR)	Audio Noise (10dB)	78.5	0.77	79.2	0.78	80.1	0.79	82.4	0.81
	Audio Noise (0dB)	74.3	0.72	76.0	0.74	77.2	0.75	80.5	0.79
	Visual Blur ($\sigma = 1.5$)	79.1	0.78	79.8	0.79	80.5	0.79	82.8	0.81
	Visual Blur ($\sigma = 3.0$)	75.6	0.74	77.1	0.75	78.0	0.76	80.9	0.79
Overall	Avg. Drop Rate ($\downarrow$)	-12.4%	-13.1%	-9.8%	-10.2%	-8.7%	-9.1%	-5.8%	-6.3%

TABLE III
ABLATION STUDY RESULTS

Configuration	Hypergraph	Contrastive	Acc (%)	F1 (macro)	Time (s/epoch)
HyperGCN + CE only	Yes	No	79.8	0.77	42.3
Standard GCN + Contrastive	No	Yes	80.5	0.78	40.9
Full HCLF (HyperGCN + Contrastive)	Yes	Yes	83.7	0.82	44.8
w/o Both	No	No	76.2	0.74	36.1

C. Fusion, Classification, and Training Details

Information propagation is performed through a Hypergraph Convolutional Network (HyperGCN), defined as:

$$H^{(l+1)} = \sigma \left(\sum_{e \in E_v} W_e \cdot H_e^{(l)} \right), \quad (4)$$

where $H^{(l)}$ denotes node representations at layer l , E_v are the hyperedges incident to node v , and σ is the activation function (e.g., ReLU).

The fused representation is obtained via attention-based weighting:

$$\text{Fused} = \sum_m \alpha_m \cdot \text{Rep}_m, \quad \alpha_m = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right), \quad (5)$$

where Q, K are query and key matrices from modality representations, and d is the dimension.

IV. EXPERIMENTS

A. Dataset and Evaluation Metrics

We evaluate on the CMU-MOSEI dataset, which includes text, audio, and visual data across 7 emotion categories. The data is split into training (1600 samples), validation (200), and test (200) sets. Evaluation metrics are weighted Accuracy, macro F1-score, Precision, Recall, and ROC-AUC. The annotated video segments provide comprehensive coverage of conversational emotions. Preprocessing balances the classes and standardizes features for fair comparison.

B. Experiments Results

HCLF demonstrates superior performance on the test set, as shown in Fig. 2. Detailed comparisons in Table I include recent SOTA models (MISA, MuT, MMIM), with HCLF showing clear gains, especially for subtle emotions like 'Disgust' and 'Fear' where high-order interaction modeling is most beneficial. Robustness tests under modality-missing

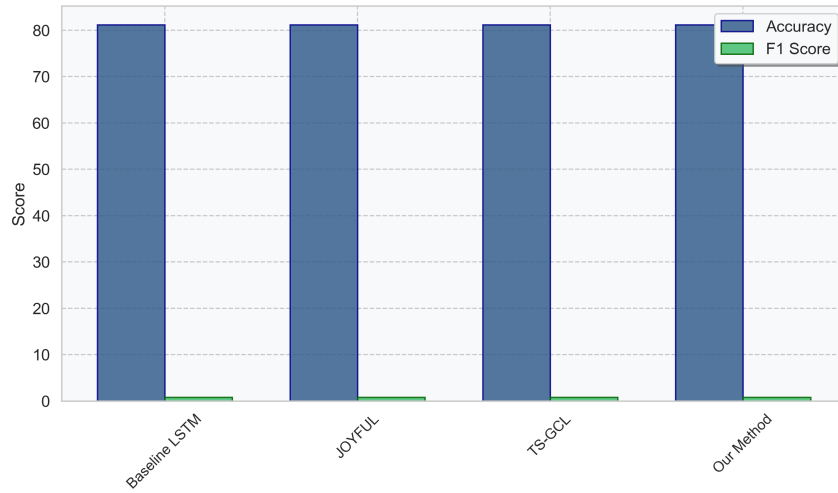


Fig. 2. Accuracy and F1 Score Comparison. Our method achieves 83.7% Accuracy, surpassing baselines including recent SOTA models (MISA, Mult, MMIM).

and noise-injection settings (Table II) confirm that HCLF maintains higher stability than baselines; for instance, with missing visual data, HCLF’s accuracy drops only 1.0%, versus 2.5–3.%for MISA and JOYFUL. Ablation studies (Table III) verify that both the hypergraph structure and contrastive learning contribute synergistically to performance.

V. CONCLUSION

The proposed Hypergraph Contrastive Learning Framework (HCLF) achieves state-of-the-art performance on the CMU-MOSEI dataset by uniquely integrating hypergraph neural networks to model high-order modality interactions with contrastive learning for semantic alignment. The results consistently demonstrate its superiority over existing baselines, confirming the effectiveness of this approach in handling multimodal complexities, especially for nuanced emotions. Future work will focus on extending the framework to asynchronous data streams, integrating physiological signals, and exploring privacy-preserving methods like federated learning.

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